Date: 20260501
Title: Medium Preparation
Purpose: Prepare medium for *Bacillus Subtilis* electroporation
Procedure:
A.Growth Medium (GM medium) :

1. Prepare medium in 500 mL flask

substance	amount
LB broth powder	6.25 g
D - sorbitol (final concentration: 0.5 M)	22.77 g
Deionized water	Add to 250 mL

2. Stir on the heat plate until dissolved
3. Autoclave for 20 min

B. Electroporation buffer (EP buffer)

1. Prepare medium in 500 mL flask

substance	amount	Final concentration
D - sorbitol	22.77 g	0.5 M
D - mannitol	22.77 g	0.5 M
Glycerol	25 mL	10 %
Deionized water	Add to 250 mL	

2. Stir on the heat plate until dissolved
3. Autoclave for 20 min

Done by Chars, recorded by Alex

Date: 20260502
Title: Plasmid Transformation
Purpose: Pick up single colony of pNW33N-L2 (sfGFP) C1 and G1 for sequencing
Procedure :

1. Prepare the following substances in two separate microtubes:

	Microtube #1	Microtube #2
DH5α	30 μL	30 μL
pNW33N-L2 (sfGFP) #C1	2 μL	
pNW33N-L2 (sfGFP) #G1		2 μL

2. Incubate the competent cells on ice for 20 minutes.
3. Heat shock at 42 ℃ for 45 seconds
4. Put them back on ice immediately for 3 minutes
5. Transfer 30 μL of the transformants onto each of two LB agar plates containing chloramphenicol.
6. Spread evenly on agar plate
7. Place the two agar plates in the incubator for 37 ℃ overnight

Done by Sophie H. recorded by Sophie H.

Date: 20260503
Title: Pick up single colony of sfGFP #G1
Purpose: To amplify the pNW33N-L2 (sfGFP) G1 clone from a bacterial colony on an LB agar plate
Procedure:

1. Place 5 ml LB broth into each 15ml of falcon tube (#1, #2, #3, and #4)
2. Add 20 µL chloramphenicol in each falcon tubes
3. Pick four individual colonies of the pNW33N-L2 (sfGFP) G1 clone and inoculate each colony into a separate tube (#1 – #4).
4. Place the four falcon tubes in the incubator and incubate at 37 ℃ overnight with shaking at 200 rpm

Done by Sophie H, Recorded by Sophie H.

Date: 20260504
Title: Miniprep & Glycerol stock
Purpose: Isolate the pNW33N-sfGFP #G plasmid and store the glycerol stock for long-term storage of plasmids
Procedure:

1. Collect 4 mL of cells by centrifuge at 7000 g for 1 minute in 1.5 mL tubes
2. Mix 0.5 mL culture medium with 50 % glycerol into the screw-top tubes:
    a. Tube 1 x 2
    b. Tube 2 x 2
    c. Tube 3 x 2
    d. Tube 4 x 2
3. The tubes were then freezed at -80 ℃
4. Add 200 µL MX1 buffer and resuspend the cells by pipetting
5. Add 250 µL MX2 buffer and mix the solution by gently inverted the tubes 8 times
6. Add 350 µL MX3 buffer and mix it by inverted 8 times
7. Centrifuge the tubes at 23444 g for 5 minutes
8. Transfer the supernatant into the column, centrifuge at 7000 g, 1 min and discard the flow through
9. Add 500 µL WN buffer and centrifuge at 7000 g for 1 min and discord the flow though
10. Discard the flow through
11. Add 700 µL WS buffer and centrifuge at 7000 g for 1 min
12. Discard the flow through
13. Centrifuge the column at 23444 g for 5 min
14. Add 50 µL EB into the center of membrane
15. Place at RT for 20 min
16. Centigrade at 23444 g for 2 min
17. Measure the concentration

Done by Gabriel

Date: 20260504
Title: electroporation competent B.sub 168 preparation
Purpose: To make the electrocompetent cells for electroporation transformation
Procedure:

1. Dilute an overnight culture of B.sub 168 16-fole in growth medium and grow at 37 ℃ to 0.600 of 0.7 (5mL +313 μL B.sub)
   Growth medium : LB medium containing 0.5M sorbitol
2. Cool the cells on ice for 10 min with shake
3. Harvest the cells by centrifugation at 4 ℃ and 5000 g for 5 min
4. Wash the cells by adding 1 mL ice-cold electroporation medium two times
   Electroporation medium : 0.5 M sorbitol , 0.5 M mannitol and 10 % glycerol
5. Suspend the cell in 1/40 of the culture volume (5mL) of electroporation medium (5mLX1/40=125μL EP buffer)
6. Stored the cells at -80 ℃

Done by Gabriel

Date:20260506
Title: Prepare Salinity Agar Plate
Purpose: Prepare salinity plate for testing trehalose rescue experiment
Procedure:

1. Prepare mixture through following the recipe below:

	Control	50 mM NaCl	50 mM NaCl with 1 mM trehalose	100 mM NaCl	100 mM NaCl with 1 mM trehalose
1/2 MS Medium	2.37 g	1.975 g	1.975 g	1.975 g	1.975 g
NaCl	–	0.1461 g	0.1461 g	0.2922 g	0.2922 g
100 mM trehalose	–	–	500 µL	–	500 µL
Deionized water	To 60 mL	To 50 mL	To 50 mL	to 50 mL	to 50 mL

2. Autoclave in 127 ℃ for 20 minutes
3. Dispense in 10 cm petri dish and wait until it solidify
4. Sow 9 arabidopsis seeds in each plate
5. Vernalization at 4 ℃, and wrapped with aluminium foil to avoid from light

Done by Chloe, supervised by Chars

Date: 20260507
Title: Miniprep
Purpose: Prepare plasmid pWN33N-L2 (sfGFP) for *Bacillus Subtilis* electroporation
Procedure:

1. Collected the cells from 20 mL TB medium liquid culture by centrifuge at 23444 g for 1 min until collected the liquid culture in 4 x 2 mL micro tube.
2. Resuspend in 250 µL Buffer P1. Note: We used Qiugen miniprep instead of Viogen miniprep.
3. Add 250 µL Buffer P2 and invert the tube 10 times.
4. Add 350 µL Buffer N3 and invert the tube 10 times.
5. Centrifuge at 23444 g for 10 min.
6. Transfer 800 µL supernatant to column. Repeat this step until all of the supernatant is collected.
7. Centrifuge at 17900 g for 1 min.
8. Remove the flow-through. Add 500 µL Buffer PB.
9. Centrifuge at 17900 g for 1 min. Remove the flow-through.
10. Add 750 µL Buffer PE.
11. Centrifuge at 17900 g for 1 min. Remove the flow-through.
12. Centrifuge the empty column for 1 min.
13. Transfer the column to a new 1.5 mL micro tube.
14. Add 50 µL nuclease-free water (~65 °C). Incubate at RT for 5 min.
15. Centrifuge at 17900 g for 3 min.
16. Measure the concentration by Biodrop.

	Concentration (ng/µL)	A260/230	A260/280
pNW33N-L2 (sfGFP) #A	56	2.24	1.867
pNW33N-L2 (sfGFP) #B	71	2.44	1.919

Done by Chars, Recorded by Sophie L

Date:20260507
Title:Prepare recovery medium
Purpose: Prepare recovery medium for *Bacillus Subtilis* electroporation
Procedure:

1. Prepare the medium by following the recipe below:

	Weight (g)	Final concentration
SOC broth powder	1.585	1 X
D-sorbitol	4.55	0.5 M

D-mannitol	3.46	0.38 M
Deionized water	Top up to 50 mL	

2. Filter through a 0.22 μm filter

Done by Chloe & Sophie H, supervised by Chars

Date: 20260509
Title: Prepare 1/2 MS agar (30 mL)
Purpose: Prepare agar for hydroponics cultivation of *Arabidopsis* Procedure:

1. Prepare the mixture through the following recipe:

	Weight (g)
1/2 MS Medium	0.069
ddH_2O	30 ml

2. Add stir bar into mixture and place the beaker with the mixture on the heat plate.
3. Cover the beaker with aluminum foil and tape a piece of autoclave tape.
4. Autoclave for 20 min at 127 °C.
5. Turn on the Biological Safety Cabinet.
6. Lift sash to operating height.
7. Wipe the working space with 70 % ethanol.
8. Spray Petri dishes, micropipette, and beaker (contain lids) with 70 % ethanol and put into the cabinet.
9. Exposure of UV for 10 mins, with sash closed.
10. Raise the sash to the certified operating height.
11. Spray hands with 70 % ethanol and place lids in each dish.
12. When the autoclave ends, waterbath the agar medium to cool down.
13. Micropipette 200 µL into each cap.
    a. Ensure all caps are placed flat on the dish.
14. Finish 3 plates of caps.

Done by Abby T, Supervised by Gina, Recorded by Sophie L

Date: 20260510
Title: Electroporation
Purpose: Testing the condition of electroporation for *Bacillus Subtilis*
Procedure:

1. Prewarm SOC- recovery Medium (SOC-RM) in 37 °C and pre-chill cuvette
2. Mix pNW33N-L2(sfGFP) (54 µg/ml) 190 µl with electrocompetent cell 90 µl
3. Transfer the plasmid–competent cell mixture into pre-chill 0.2 cm cuvette Setup electroporator

	Voltage (kV)	Capacity (µF)	Resistance (Ω)	TC (ms)	Notes
Trial 1	0/2.0	25	∞	0	Arc
Trial 2	0/1.8	25	∞	0	Arc
Trial 3	1.995/2.0	25	∞	6553.5	Dilute 1/2
Trial 4	2.195/2.2	25	300	6.7	
Trial 5	2.091/2.1	25	300	6.7	

4. Add 1 ml of SOC-RM into cuvette and transfer to 15 mL tube
5. Incubate at 37 °C and 200 rpm for 4 hours
6. Trial 3,4,5 were used for the subsequent experiment
7. Prepare three LB agar plates with chloramphenicol and three without chloramphenicol.
8. Prepared the diluted samples 1:1,000,000 for each one
9. The remaining samples (original) were centrifuged at 5000 g for 5 minutes
10. The dilute samples were spread on LB plate
11. The original samples were spread on LB plate with chloramphenicol
12. Incubate the plate at 37 °C overnight

Done by Gabriel

Date: 20260511
Title:Preparation of *Bacillus subtilis* competent cells
Purpose: Prepare electrocompetent *Bacillus subtilis* cells that can efficiently take up foreign DNA during electroporation.
Procedure:

1. Dilute an overnight culture of *Bacillus Subtilis* in 16 fold in growth medium (LB medium containing 0.5 M sorbitol) (312.5 µL of *Bacillus Subtilis*) in 5 ml Growth medium and grow at 37 °C to achieve OD600 to 0.944
2. Cool the cells on ice cold water and put on shaker for 10 minutes
3. Centrifuge at 4 °C at 5000 g for 5 mins to harvest cells (total of 4.5 mL)
4. Add 1 mL ice cold electroporation medium to resuspend the cells in 1.5 mL microtube, centrifuge 5000 g for 1 minute and discard supernatant
5. Repeat step 4 for a total of 4 times to wash cells
6. Suspend the cells in 1/40 of the culture volume of the electroporation solution (0.1125 mL) with a cell concentration of 1-1.3 x 10^10 cfu/mL
7. Store competent cells at -80 °C

Problems encountered: Step 4 produced small pellets so centrifuged for 5 mins instead of 1 minute

Done by Chloe, documented by Olivia Lin

Date: 20260512
Title: Purification of plasmids
Purpose: Check if *Bacillus subtilis* electroporation (transformation) was successful
Procedure:
I. Plasmid preparation(pNW33N-L2 (sfGFP)

1. Pick up single colonies and transfer each to 5 ml of LB medium
2. Add 5 µL of chloramphenicol (25 mg/ml) to each
3. Let the culture incubate for 25 °C overnight

II. Purification of plasmids (applies to all 7 plasmids)

1. Transfer 2 ml of the overnight culture to a 2ml microtube
2. Centrifuge at 16000 g for 1 min
3. Discard the supernatant
4. Make the MX1 buffer mix consisted of MX1 buffer and lysozyme

4-1 Calculations for 7 tubes:
-Amount of MX1: 250 µL*9=2250 µL
-For 1mg/ml concentration of lysozyme:2250 mg/1000=2.5 mg
-For 10mg/ml concentration of lysozyme: 2.25 mg/10=0.225ml=225 µL
-Buffer: 2250 µL-225 µL=2025 µL

5. Add 200 µL of the MX1 buffer mix
6. Incubate at 37 °C for 10 mins
7. Resuspend the pellet by pipetting
8. Add 250 µL of MX2 buffer
9. Gently mix the solution by inverting the microtube 8-10 time
10. Incubate the mixture at RT for 2 min
11. Add 350 µL of MX3 buffer
12. Gently mix the solution immediately
13. Isolate the supernatant with plasmid and centrifuge it at 16000 g for 5 min
14. Transfer the supernatant to column and centrifuge at 7000 g for 30 sec (discard the flow through)
15. Add 500 µL of WN buffer, centrifuge at 7000 g for 30sec and discard the flow through
16. Add 700 µL of WS buffer, centrifuge at 7000 g for 30sec and discard the flow through
17. Centrifuge the empty column at 10000g for 3min to remove residual ethanol
18. Place the column into a new 1.5mL microtube
19. Add 50 µL of elution buffer on the center of the membrane
20. Incubate the column at RT for 3 min then centrifuge at 10000 g for 3 min
21. Measure the concentrations with Biodrop

Result:

Sample	Concentration (mg/mL)	A260/280	A260/230
#1	21	1.909	0.913
#2	18	1.800	1.125
#3	11	2.200	5.500
#4	16	2.000	2.667
#5	27	1.929	1.125
#6	16	2.000	5.333
#7	20	1.667	0.500

The DNA concentration will thus be used to calculate the reaction formula of enzyme digestion

Done by Sophie C. Recorded by Sophie H.

Date: 2026/5/12
Title: pNW33N-L2 (sfGFP) plasmid linearization
Purpose: To prepare the purified pNW33N-L2 (sfGFP) plasmids for gel electrophoresis analysis
Procedure:

1. Calculate reaction formula:

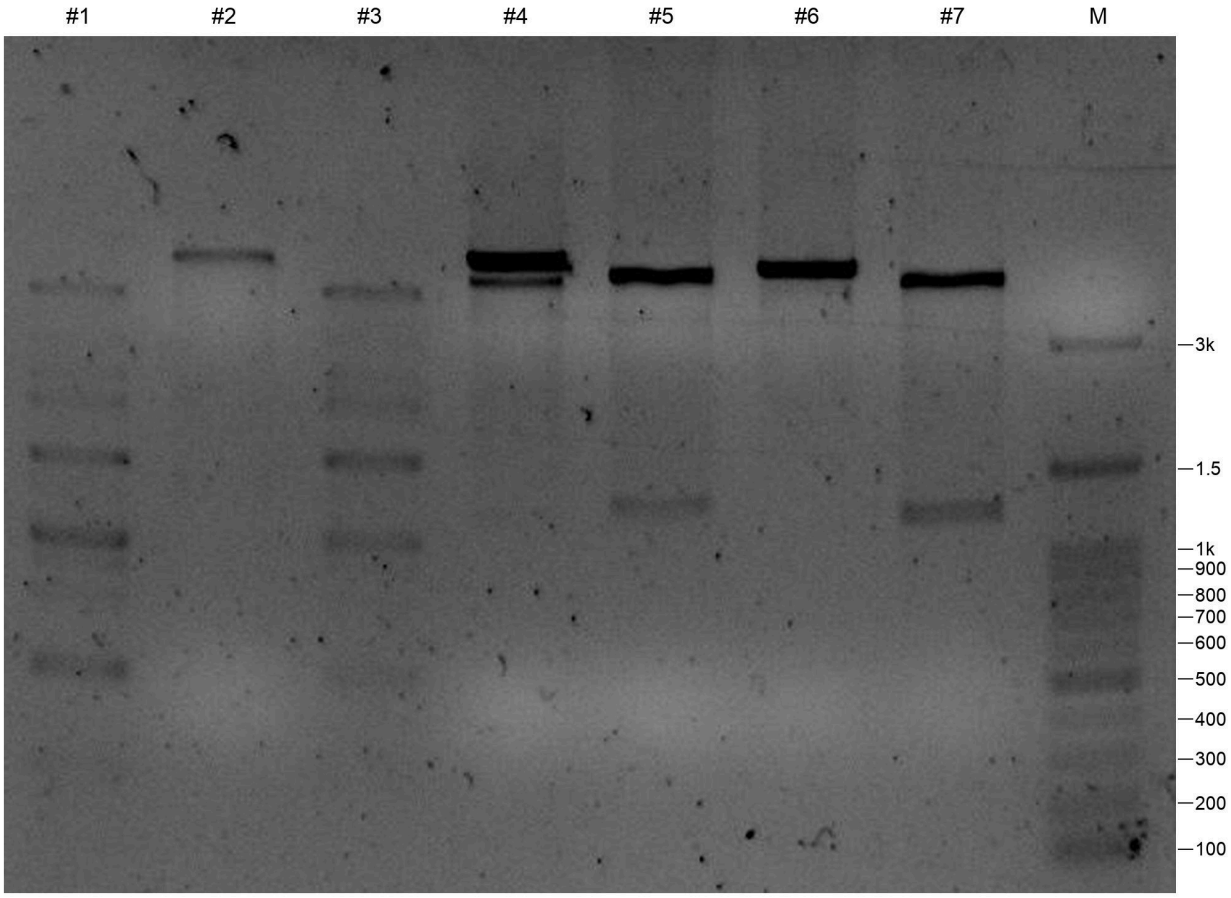
Sample	DNA	ddH_2O	EcoRI	DNA1μL	ddH_2O	BtsI
#1	11.9	5.6	0.5	11.9	2.8	1
#2	13.9	3.6	0.5	13.9	1.8	1
#3	20	0	0.5	20	0	1
#4	15.6	1.9	0.5	15.6	0.95	1
#5	9.3	8.2	0.5	9.3	4.1	1
#6	15.6	1.9	0.5	15.6	0.95	1
#7	12.5	5	0.5	12.5	2.5	1

+10X rCutSmart buffer in each tube : 2 µL

*Samples 1~7 are plasmids from 5/12's purification under the conditions of 2.2 kv, 300Ω. And the plasmid is pNW33N-L2 (sfGFP)

2. Prepare reaction by adding ddH_2O and buffer first, then add DNA and enzymes
3. Centrifuge the 7 microtubes by balancing 6 microtubes for 1st round of centrifuge and keep one microtube with the one left to balance for second round to centrifuge(each round 5 sec)
4. Put microtubes in PCR machine at 37 °C for an hour to incubate
5. Electrophoresis
    a. Add and measure 0.3 grams of agarose powder
    b. Measure 30 ml of TAE buffer in measuring cylinder
    c. Wrap flask with plastic (food) wrap on the opening
    d. Use pipette tip to poke 4~5 holes in the wrap
    e. Place flask in microwave for 30 sec (wear cotton gloves)
    f. Take out the flask and gently shake to see if the mixture is clear and all the powder has dissolved.
    g. If not, microwave for another 30 sec
    h. Assemble the tray (8 slots: 1 DNA ladder, 7 samples)
    i. Pour agarose mixture in tray
    j. Use the large tip to press the bubbles in the mixture/tray away
    k. Wait for gel to solidify and take gel into electrophoresis tray containing the TAE buffer
    l. Take samples out of the PCR machine
    m. Aliquot 2 µL of 6 X loading dye onto the parafilm for each sample. Pipette
    n. Pipette and mix 10 µL of each sample with the dye. Then, load 10 µL of the mixture into each well.
    o. Connect the electrodes (black-black, red-red) and run the gel at 100 V , check for bubbles in the buffer and make sure the samples are moving toward the positive electrode.

Results:



Gel electrophoresis of Linearized pNW33N-L2(sfGFP) plasmid

Lane	Sample	Treatment
1	pNW33N-L2(sfGFP)	Enzyme digestion for plasmid linearization
2	pNW33N-L2(sfGFP)	Enzyme digestion for plasmid linearization
3	pNW33N-L2(sfGFP)	Enzyme digestion for plasmid linearization
4	pNW33N-L2(sfGFP)	Enzyme digestion for plasmid linearization
5	pNW33N-L2(sfGFP)	Enzyme digestion for plasmid linearization
6	Not specified	Not specified
7	Not specified	Not specified
M	1 kb DNA Ladder	

Done by Abigail, Abby Tsai, Recorded by Sophie H.

Date: 20260513
Title: Colony PCR
Purpose: Verify the results of the electroporation
Procedure:

1. Prepare the master mix

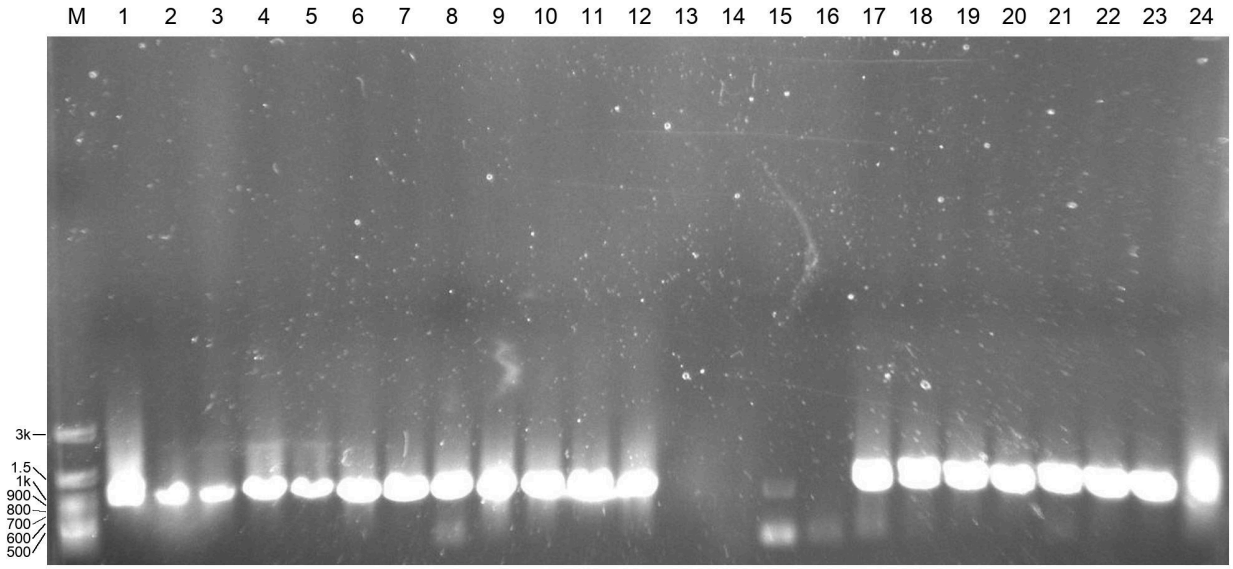
	1 rxn	25 rxn	
Forward	1 µL	25 µL	FW primer : BioBrick prefix
Reverse Primer	1 µL	25 µL	RV primer: BioBrick suffix
Q-Amp Master Mix	10 µL	250 µL	
ddH_2O	8 µL	200 µL	

2. Dispense reaction to 8-strip PCR tube
3. Pick up the colonies from the 2.0 kv, 2.1 kV, 2.2 kV condition plate
4. React in the following thermal cycle

95 ℃	5 min	1 cycle
95 ℃	30 sec	35 cycles
55 ℃	30 sec	
72 ℃	1 min	
72 ℃	5 min	1 cycle

5. Prepare a 1 % agarose gel
6. Resolved PCR product in 1 % agarose gel

Result:



Done by Chloe, Olivia Lin, Abigail
Documented by Olivia Lin

Date: 20260514
Title: Miniprep
Purpose: Prepare the shuttle vector pNW33N-L2 (sfGFP) for electroporation
Procedure:

1. Transfer 1500 µL of overnight liquid culture inoculated from colonies on the streak plates used for the colony (Done in 5/13) to each tube (total 8 tubes)
2. Centrifuge for 3 min at 6800 g, 22˚C
3. Aspirate supernatant from 8 tubes with micropipette
4. Add the rest of the liquid culture
5. Centrifuge at 22˚C 23944 g for 1 minute
6. Collect supernatant transfer to column
7. Centrifuge at 17900 g for 30 seconds
8. Repeat this step, until collect all the supernatant
9. Add 500 µL PB buffer and centrifuge at 17900 for 30 seconds.
10. Discard the flow-through at 750 µL PE buffer
11. Centrifuge at 17900 g for 3 minutes
12. Discard the flow through and centrifuge at 17900 g for 3 minutes.
13. Transfer the column to new 1.5 ml microtube
14. Add 50 µL nuclease free water at 65˚C
15. Incubate at RT for 3 minutes
16. Centrifuge at 17900 g for 3 minutes
17. Measure the concentration

	Concentration (ng/µL)	A260/230	A220/280
pNW33N-L2 (sfGFP) #1	53.0	1.606	1.893
pNW33N-L2 (sfGFP) #2	142.0	2.152	1.919

Done by Chloe, Olivia Lin, Ryan Yuan

Date: 20260514
Title: Preparing NaCl or Trehalose containing agar plate -> remake 20260506's salinity plate
Purpose: Prepare the mixture for new salinity plates
Procedure:

1. Prepare the mixture by following the recipe below

	control	500 mM NaCl	50 mM NaCl / 0.5 mM trehalose	100 mM NaCl	100 mM NaCl w/ 0.5 mM trehalose
1/2 MS medium	1.975 g	1.975 g	1.975 g	1.975 g	1.975 g

NaCl		0.1461g	0.1461g	0.2422 g	0.2922 g
100mM trehalose,			25 µL		250 µL
deionized water	50 ml	50 ml	50 ml	50 ml	50 ml

2. Autoclave at 127˚C for 20 mins
3. Dispense in 9 cm Petri dish, dry for 1 hour
4. Place 9 Arabidopsis seeds in each plate in a grid pattern
5. Vernalization at 4˚C, avoid light
6. Add 30 ml of deionized water and stir with hot plate before adding the rest to 50 ml
7. Autoclave at 127˚C for 20 mins
8. Arrange the concentration to 100 ml and 150 ml
9. Add a control with trehalose and no sodium chloride

Done by Abby Kao, Abigail, Audrey H, Naomi

Date: 20260506
Title: Preparing Hydroponics Medium
Purpose: Prepare MS medium with adjusted pH levels for hydroponic system
Procedure:

1. Prepare the mixture by following the recipe below.
    a. 1/2 MS medium : 4.6 g / 2L
    b. ddH_2O : 2 L
2. Pour about 900 mL of ddH_2O inside the beaker and stir it with the MS medium powder using a magnetic stir bar.
3. Before using the pH meter, use pH 4 and pH 7 buffer solutions to do two-point calibration.
4. Use the pH meter to determine the pH of the medium. Adjust pH by adding KCl. Record the before and after pH results.
5. Fill the rest of the ddH_2O into the beaker.
6. Distribute 2 L of the mixture to five 500 mL flasks, each flask is filled with 400 mL of medium mixture.
7. Cover up the opening of all flasks with foil and stick the autoclave indicator tape on the foil.
8. Pour 1100 mL of water to the autoclave and autoclave the five flasks for 20 minutes.
9. Use tap water to cool down each flask.
10. After cooling, fill the hydroponics box with medium.
11. Transfer the plant seed to the hydroponic box.

Result:
pH results of mixture:

- Before pH result: pH 4.05
- After pH result: pH 5.62

Done by Naomi Lin, Joshua Hong, Sara Chen, Sophie Huang
Recorded by Naomi Lin, Sara Chen

Date:20260517
Title: Electrophoration
Purpose: Testing the condition for electroporation - remake 5/10 to increase colony number
Procedure:

1. Prewarm SOC-recovery medium (SOC-RM) in 37 °C and precold cuvette.
2. Mix pNW33N-L2 (sfGFP) (142 ng/mL) 7 µL and electrocompetent cell 93 µL
3. Pre-chill 0.2 cm cuvette with plasmid-competent cell mixture
4. Set up electroporator

	Voltage (kV)	Capacity (µF)	Resistance (Ω)	TC (ms)
Trial 1	1.977/2.0	25	200	5.05
Trial 2	2.074/2.1	25	200	
Trial 3	2.182/2.2	25	200	4.6

5. Add 1 ml of SOC-RM to each tube, transfer to 15 mL tubes. Don't fasten the tube completely, tape it to fix the lid of the tube.
6. Incubate at 37˚C, 200 rpm for 4 hrs
7. Put the LB plate in the incubator
8. Dilute plasmid with LB medium with the ratio of 1:100,000
    a. First dilute 1:1000 (1 µL:999 µL), then 1:100 (1 µL:99 µL)
9. Use the applicator stick to spread the diluted plasmid solution over LB plate
10. Take the rest of the 1:100 mixture and centrifuge it at 5000 g for 5 minutes.
11. Discard the supernatant and leave the pellet.
12. Resuspend the tubes.
13. Spread 100 µL of the LB culture onto a chloramphenicol-containing LB agar plate.
14. Incubate overnight

Done by Sara Chen, Sophie Huang, Chloe Wu, Abigail

Date: 20260517
Title: Prepare 1 Liter of LB Agar Plate
Purpose:Prepare LB agar plate
Procedure:

1. Clean the flasks with tap water followed by followed by rinsing ddH_2O, add magnetic stir bar
2. Measure 25 g of LB, 15 g of Agar (3.75 g / 250 mL), and add 1000 ml of ddH_2O into the flask
3. Distribute 500 mL to two separate 500 mL flasks.
4. Cover each flask opening with foil and stick autoclave indicator tape on top of the foil. Autoclave for 20 minutes.

Done by Chloe W, Ryan Wei, Ryan Yuan, Abigail, Naomi, Phoebe Chen, Alex

Recorded by Naomi

Date: 20260518
Title: Remaking *Bacillus subtilis* Competent Cell
Purpose: To prepare for electroporation
Procedure:

1. Dilute the original bacterial culture at a 1:16 ratio and distribute it into six 10-mL tubes, for a total volume of 60 mL.
2. Grow the bacteria culture at 37 °C to an OD600 of 0.944
3. Take 2 ml of the culture for testing absorbance
4. Repeat step 1 & 2 until the culture's absorbance OD600 is 0.85-0.95
    a. Trial 1: OD = 0.67 (after 2 hours)
    b. Trial 2: OD = 1.205 (after 1 hour; could adjust time to 30 mins)
5. Cool the diluted tubes for 10 minutes in ice bath
6. Place the ice bath on the shaker to cool the solution quickly
7. Centrifuge 5000 g at 4 °C for 5 minutes
8. Prepare and continue lab work under aseptic operating conditions
9. Add a total of 4.8 ml cells into a microtube
10. Add 1ml of cold Electroporation Buffer
11. Pour out the supernatant
12. Add 1 ml of cold Electroporation Buffer
13. Remove the supernatant
14. Resuspend the cells with a 40:1 ratio of 4.8 ml (add 1.2 ml or 0.12 ml)
15. Place the microtube in the -80 °C fridge overnight

Done by: Chloe , Sara C, Sophie H, Abigail L, Renee K, and Ethan C

Date: 20260519
Title: Electroporation
Purpose: Testing the conditions by which has the best transformation efficiency on *Bacillus subtilis*
Procedure:
1. Rewarm the recover medium at 37 °C
2. Mix pNW33N-L2 (sfGFP) (142 ng/ml) 7 microliters to 93 microliters of competent cells (5/18)
3. Pre-chill 0.2cm cuvette with plasmid competent cell mixture
4. Set up electroporator

	Voltage (kv)	Capacity ((μF)	Resistance (Ω)	TC (ms)
Trial 1	2.0	25	200	
Trial 2	2.1	25	200	4.0
Trial 3	2.2	25	200	3.9

5. Immediately add 1mL SOC-RM
6. Transferred all medium and plasmids (liquid content) to the 15 mL tube
7. Tape the lid while not completely fasten the tube as it is airtight
8. Incubate at 37˚C, 209 rpm for 3 hrs
9. Put the agar plate into the incubator.
10. Dilute the plasmid using LB medium with a ratio of 1: 1000 and then 1: 100 thus ending with a dilution of 1:1,000,000.
→ for 1:1000 use 1 μL: 999 μL
→ for 1:100 use 1 μL: 99 μL
11. Use application stick to spread the diluted plasmid solution over the LB plates
12. Take the rest to the 1:100 mixture and centrifuge it at 5000 g for 5 minutes
→ remove the supernatant and add 100 microliters of LB medium to resuspend pellet & repeat step 11 again with new LB plate
Done by Chloe W, Olivia Lin, Abigail, Abby T

Date: 20260519
Title: Medium Preparation
Purpose: Prepare growth medium for *Bacillus subtilis*
Procedure:
1. Prepare the medium in 500 mL flask
   - LB broth: 6.25 g
   - D-sorbital: 22.77 g → concentration to 0.5 M
   - Deionized water: Add 250 ml
2. Stir on the heat plate until fully dissolved
3. Autoclave for 20 minutes
Done By Chloe W, Olivia Lin, Abigail, Abby T.

Date: 20260519
Title: Electroporation competent cell (*Bacillus subtilis*) preparation
Purpose: To prepare the competent cell for electroporation
Procedure:

1. Dilute the overnight culture of *Bacillus subtilis* to 16-fold in medium (GM medium).
2. Grow the *Bacillus subtilis* in 37 °C, 200 rpm to OD600 of 0.915.
3. Cool the cells on ice for 10 min with shake.
4. Harvest the cells by centrifugal ion at 4 °C and 5000 g for 5 min.
5. Wash the cells twice with EP buffer (add 1 ml and discard it).
6. Resuspend the cells in 1/40 volume of the culture volume with EP buffer.
7. Store the cells in -80 °C.

Done by Gabriel, Recorded by Sophie L

Date:20260520
Title: Miniprep
Purpose: Isolate and purify small amounts of plasmid DNA from a small culture of *Bacillus subtilis*
Procedure:

1. Pellet 1.8 mL bacterial overnight culture by centrifugation at >8000 rpm (6800 g) for 3 minutes at room temperature
2. Resuspend pelleted bacterial cells in 250 µLl Buffer P1
3. Add 250 µL Buffer P2 and mix by inverting the tube 4-6 times until the solution becomes clear (DO NOT PIPETTE!!)
4. Add 350 µL Buffer N3 & mix by inverting the tube 4-6 times
5. Centrifuge for 5 minutes at 23444 g
6. Apply 800 µL supernatant from step 5 to column by pipetting. Centrifuge for 30 seconds & discard follow through. Repeat two times until the original solution is completely used up
7. Wash by adding 0.5 mL Buffer PB. centrifuge for 30 seconds and discard the follow through
8. Wash by adding 0.75 mL Buffer PE. centrifuge for 30 seconds and discard follow through
9. Centrifuge for 1 min to remove residual wash buffer
10. Place column in 1.5 mL microcentrifuge tube and add 50 µL nuclease free water to the center of the column. Let it stand for 1 minute and then centrifuge it for 1 minute.

Results:

	Concentration (ng/ul)	A260/A280	A260/A280
Sample 1	216	2.455	1.895
Sample 2	291.9	1.508	1.865

Done by Sophie Chen and Chloe. Recorded by Chloe

Date:20260520
Title: LB Agar plate
Purpose: Prepare agar plates for electroporation
Procedure:

1. Formula as below:

Components	Mass
Broth Powder	6.25 g
Agar powder	3.75 g
ddH_2O	250 ml

2. Add one magnetic stirring bar to the flask
3. Place the flask containing the mixture on the hot plate
4. Remove the stirring bar from the flask
5. Cover the flask with aluminum foil and apply an indicator tape to ensure the mixture is successfully autoclaved
6. Place the flask in autoclave
7. Place the new petri dishes in BSC and turn on the UV light
8. Set up water bath at 55 °C
9. Disinfect the working surface of the BSC and all equipment by spraying them with 70 % ethanol.
10. Add 500 µL of antibiotics (ampicillin, chloramphenicol) into each corresponding flasks
11. Use serological pipette to add 10-15 ml of melting LB agar into the petri dish, avoid overfilling
12. Arrange the petri dishes in stacks inside the cabinet
13. Let the agar plates dry for at least 1 hour in the BSC

Done by: Abby T and Abigail, Recorded by: Abby T

Date: 20260521
Title: Electroporation
Purpose: Testing the conditions by which has the best transformation efficiency on *Bacillus subtilis.*
Procedure:

1. Prewarm the recovery medium at 37 °C
2. Mix pNW33N-L2 (sfGFP) (216 µg/mL 4.6 µL, 95.4 µL electrocompetent cells)
3. Pre-chill 0. 2 cm cuvette with plasmid- competent cell mixture
4. Set up electroporator

	Voltage (kv)	Capacity ((µF)	Resistance (Ω)	TC (ms)
Trial 1	1.987/2.0	25	300	6.5
Trial 2	2.092/2.1	25	300	6.7
Trial 3	2.195/2.2	25	300	6.8

5. Immediately add 1 mL SOC-RM to the cuvette
6. Transfer all medium and plasmid (liquid content) to the 15 ml tube
7. Tape the lid while not completely fastening the tube as it is air tight
8. Incubate at 37 °C 209 pm for 3 hrs

Done by Sophie L. Supervised by Chars

9. Put the Agar Plate into the incubator (for pre-warming)
10. dilute the plasmid using LB medium with a ratio of 1:1000 and then 1:100 thus ending with a dilution of 1:1000,000

→ for 1:1000 (1 µL: 999 µL)
→ for 1:100 (1 µL: 99 µL)

11. Use application stick to spread the diluted plasmid solution over the LB plates
12. Take the rest of the 1:100 mixture and centrifuge at 5000 g for 5 minutes
13. remove the supernatant and add 100 µL of LB medium to resuspend pellet
14. repeat step 1 again with new LB plate

Done by Ryan Yuan, Chloe W
Supervised by Olivia Lin

Date: 20260522
Title: 1/2 MS Medium preparation
Purpose: Preparation for 5 % growth medium
Procedure:

1. Formula as below:

- 1/2 MS medium powder
    - Mass: 1.975 g
- Distilled water
    - Volume: 50 ml

2. Use a hot plate to stir until no sediments left in the solution
3. Wrap the neck with aluminium foil and stuck a piece of indicator tap on top
4. Autoclave for 20 minutes at 150 °C
5. Pour the agar solution 1ml into the 1.5 ml tube lid with a pre-punched 2 mm hole.
6. Wait for the agar to solidify
7. Place one Arabidopsis seed into the 2 mm hole of each of the 40 eppendorf tube caps.

Done by Abby K, Ethan Chang, Olivia Lin, Sophie H

Date: 20260523
Title: Electroporation
Purpose: Testing the conditions by which has the best transfer efficiency on B. subtilis
Procedure:

1. Prewarm the recovery medium at 37 °C
2. Mix 4.63 µL of pNW33N-L2 (sfGFP) (concentration 216 µg/ml) and 95.37 µL of competent cell
3. Pre-chill 0.2 cm cuvette with plasmid-competent cell mixture
4. Set up electroporator with the following settings:

	Voltage (kV)	Capacity (µF)	Resistance (Ω)	TC (ms)
Trial 1	2.089/2.1	25	300	5.1
Trial 2	2.280/2.3	25	300	4.4
Trial 3	2.486/2.5	25	300	3.9

5. Immediately add 1ml of SOC-RM to the cuvette
6. Transform all medium and plasmid to 15 ml tube
7. Tape the lid while not completely fastening it airtight
8. Incubate at 37 °C at 200 rpm for 3 hours

Done by Naomi, Sara, Phoebe, Ethan C, Sophie C, Sophia Y

9. Put the agar plate into the incubator to prewarm
10. Dilute the plasmid using LB medium with a ratio of 1:1000 then 1:100 thus ending with a final dilution of 1:1,000,000
    a. For 1:1000 use a ratio 1 µL : 999 µL
    b. For 1:100 use a ratio of 1 µL to 99 µL
11. Use application stick to spread the diluted plasmid solution over the LB plates
12. Take the rest of the 1:100 mixture and centrifuge at 5000 g for 5 minutes
13. Remove the supernatant and add 1000 µL of LB medium to resuspend the pellet
14. Repeat step 11 again with LB plates containing chloramphenicol

Done by Olivia Lin, Phoebe, Ethan Chang, recorded by Olivia Lin, Naomi Lin, Sarah Chou

Problems encountered:

- someone doing the first part of the experiment forgot to Label the voltage values onto the 15ml tube so we are not sure which is 2.3 kV and which is 2.5 kV
- The 2.1 kV solution (for some reason) had half the volume of the other two resulting in the same dilution factor but using 0.5 µL : 499.5 µL for 1:1000 ratio and 0.5 µL : 49.5 µL for 1:100 ratio

Date: 20260524
Title: Agar Plate Preparation
Purpose: Preparation for agar plate
Procedure:

1. Formula as below:

	Amount
1/2 MS Medium Powder	3.95 g
Distilled Water	100 mL
NaCl	0.5844 g

2. Use a hot plate to stir (with a stirring rod) until no sediments remain in the solution.
3. Wrap the neck of the flask with aluminum foil and stick a piece of indicator tape on top.
4. Autoclave for 20 minutes at 127 ℃.
5. Wait for the pressure to reach 0 before removing the flask.
6. While waiting for the BSC to be ready, keep the flask in the water bath to prevent solidification.
7. Pipette 20 mL of agar solution into each plate. Repeat for a total of five plates.
8. Wait for the agar to solidify.
9. Defrost the trehalose.
10. Disinfect the spatula by dipping it into fuel alcohol then heating it with a Bunsen burner.
11. Wait for the spatula to cool.
12. Pipette the respective volumes of trehalose for different molarities. Do one plate for each.

Molarity (Trehalose Concentration)	Volume (Stock: 100 mM)
0.125 mM	25 µL
0.25 mM	50 µL
0.5 mM	100 µL

13. Spread the trehalose with the spatula until the spreading is not smooth.
14. Disinfect the spatula and repeat for each molarity.

Done by Chloe W, Ethan C, and Sophia Y

Date: 20260525
Title: Agar Plate Preparation
Purpose: Preparation for agar plate
Procedure:

1. Formula as below:

1/2 MS Medium Powder	5.925 g
Distilled Water	150 mL
NaCl	0.8766 g

   Add the contents into a flask.
2. Use a hot plate to stir (with a stirring rod) until no sediments are left in the solution.
3. Wrap the neck with aluminum foil and stick a piece of indicator tape on top.
4. Autoclave for 20 minutes at 127 °C.
5. Wait for the pressure to reach 0 before retrieving the flask.
6. While waiting for the BSC to be ready, keep the flask in the water bath to prevent solidification.
7. Transfer 20 mL of agar solution into each plate. Repeat for a total of five plates.
8. Wait for the agar to solidify.
9. Thawthe trehalose.
10. Disinfect the spatula by dipping it into fuel alcohol then heating it with a Bunsen burner.
11. Wait for the spatula to cool.

Done by: Chloe, Olivia L, Sophie L, Sara C, Recorded by Sophie L

Date: 20260525
Title: Competent Cell
Purpose: Making Competent Cell for Electroporation
Procedure:

1. Dilute an overnight culture of *Bacillus subtilis* in 16 fold in growth medium (LB medium containing 0.5 M growth medium) (313 µL B. sub. In 5 ml growth medium), grow at 37 °C to an OD600 of 0.851
2. Cool the cells on ice and put the ice bucket to harvest cells
3. Centrifuge at 4 degree at 5000 g for 5 min to harvest cells
4. Add 1 µL of ice cold electroporation medium to resuspend cell in 1.5 ml microtube, then centrifuge at 5000 g for 1 minute and discard supernatant
5. Repeat 4 times to wash the cells
6. Suspend the cells in 1/40 of the culture volume of the electroporation solution (0.125 ml)
7. Store competent cell at -80 °C

Done by Chloe, Olivia Lin, Sophie Liu, Sara Chen Supervised by Gabriel

Date: 20260526
Title: Electroporation
Purpose: To test/optimal voltage for enhancing the electroporation efficiency
Procedure:

1. Mix the plasmid pNW33N-L2 (sfGFP) and the competent cell (5/25) Plasmid (13μL) + competent cell (287 μL) (216 ug/μL).
2. Transfer 100 μL to each cuvette (0.2 cm).
3. Ice bath the cuvette for 3 min.
4. Set up and start the electroporation process.

	Voltage (kV)	Capacity (μF)	Resistance (Ω)	TC (ms)
Cuvette 1	2092	25	300	7.3
Cuvette 2	2285	25	300	7.3
Cuvette 3	2493	25	300	7.2

5. Immediately add 1 mL warm recovery medium to the cuvette and transfer to the tubes.
6. Incubate at 37 °C, 200 rpm for 3 hr.
7. Prepared 3 LB plate and 3 LB plate with Chlorophenicol.
8. Prepare the dilute samples of each in 1:1000000.
9. The remaining samples were then centrifuged in 5000 g for 5 min.
10. The dilute samples were spread in the LB plate.
11. The remaining samples were spread in the LB plate with Chlor.
12. Incubate in 37 °C overnight.

Done by Gabriel, Recorded by Sophie L

Date: 20260526
Title: Seeding arabidopsis in the agar plate
Purpose: Sow seeds on agar plate
Procedure:

1. Volume

Molarity (Trehalose Concentration)	Volume (Stock: 100 mM)
0.125 mM	37.5 μL
0.25 mM	75 μL
0.5 mM	150 μL

2. Spread the difference volume of 1/2 Ms medium on the 1/2 M agar plate
3. Seeding 9 arabidopsis seeds on each agar plate, separate with equal interval

4. using aluminum foil cover it and label the seeding time and the time it nee to taken out (3 days after)
5. Putting into 4 ℃ and wait for 3 days vernalization

Done by Sarah Chou

Date: 20260526
Title: Hydroponic medium preparation
Purpose: Prepare hydroponic medium
Procedure:

1. Measure 2.3 g 1/2 MS medium in 1000 mL de-ionized water(mix with stirring bar)
2. Determine pH of the solution using a pH meter
   Before pH 4.0 / after pH 5.67
3. Divide 1000 mL solution into two 500ml of bakersAutoclave for 20 min at 127 ℃
4. Fill the hydroponic box with medium and transplant the seedlings into the box.

Done by Chloe W , Record by Sarah Chou

Date: 20260527
Title: Hydroponic Medium Preparation
Purpose: Prepare medium for Hydroponic system
Procedure:

1. Measure 1.15 g 1/2 MS medium powder into 1000 ml flask with 500 ml deionized water.*3 (medium#1, medium#2, medium#3)
2. Mix the medium with stirring bar
3. Determine the pH of the solution via using a pH probe
4. Measure the pH and adjust it with KCl

Medium	Before adjust (pH)	After adjust (pH)
#1	4.09	5.02
#2	4.06	5.64
#3	4.08	5.56

5. Autoclave pH-adjusted medium at 121 °C for 20 mins.
6. Let the three medium cool down in room temperature

Done by Sophie L, recorded by Sophie H.

Date: 20260527
Title: Miniprep
Purpose: experiment abort because unable to resuspense pellet properly
Procedure:

1. Transfer 1.8 mL TB medium to 2 ml microtube repeat 7 time
2. Centrifuge at 16000g for 1 minute *balance microtube
3. Discard the supernatant
4. repeat step 1 until collected all medium
5. Add 200 µL MX1 buffer (store at 4 ℃)
6. Resuspend the pellet by pipetting

*Experiment abort because unable to resuspend pellet properly

Done by : Sophie C , Sophie L , Sophia Y , Alex
Supervised by Venus

Date: 20260529
Title: Preparation LB agar plate
Purpose: Preparation LB agar plate
Procedure:

1. Formula as below:

	mass(g)		mass(g)
Broth powder	10	Broth powder	10
Agar powder	6	Agar powder	6
Distilled water	400 mL	Distilled water	400
		Kanamycin (50 mg/mL)	0.4 ml

2. Add broth , agar powder, distilled water in a flask
3. Stir until no sediment
4. Wrap the neck with aluminum foil and stick a piece of indicator tape on fep to ensure the mixture is successfully autocate
5. Place the flask in autocate
6. Set up water bath at 55 ℃
7. While waiting for the BSC to be ready , keep the flask in the water bath to prevent solidification
8. Add 400 µL kanamycin
9. Using pipette add to aliquot 20 mL of agar solution into each plate , repeat for all plate
10. Wait for agar to solidify

Done by Sarah C , Sophie H , Olivia L, Ryan Y , Sophia Y

Date: 20260530
Title: Agar Plate Preparation
Purpose: Preparation of agar plates for *Arabidopsis* planting
Procedure:

1. Formula as below:

1/2 MS Medium Powder	0.5925 g
Distilled Water	15 mL
NaCl	0.13149 g

2. Use a hot plate to stir (with a stirring rod) until no sediments are left in the solution.
3. Wrap the neck with aluminum foil and stick a piece of indicator tape on top.
4. Autoclave for 20 minutes at 127 °C.
5. Wait for the pressure to reach 0 before retrieving the flask.
6. While waiting for BSC, keep the flask in the water bath to prevent solidification.
7. Pipette 500 µL agar solution into each well of 24-well plates. Repeat for 2 well plates.
8. Wait for agar to solidify.
9. Repeat previous steps to prepare another spare.
10. Thawthe trehalose.
11. Add five 2.5 mm glass beads into each well from rows B–D (inside the BSC).
12. Pipette respective volumes of trehalose for different molarities (50 µL per well).

	Molarity (Trehalose Concentration)
Row B	0.25
Row C	0.50
Row D	0.75

13. Shake the well plates until trehalose dries.

Done by: Sophie L, Chloe, Sara Chen, Recorded by Sophie L

Date: 20260530
Title: LB Medium Preparation
Purpose: Prepare Bioreactor Medium to culture *Bacillus Subtilis*
Procedure:

1. Formula as below:

LB broth powder	12.5 g

ddH_2O	500 mL

2. Add one magnetic stirring rod to the flask.
3. Place the flask containing the mixture on the hot plate, and stir until dissolved.
4. Remove the stirring rod from the flask.
5. Cover the flask with aluminum foil and apply an indicator tape to ensure the mixture is successfully autoclaved.
6. Place the flask in autoclave.

Done by: Sophie L, Chloe, Recorded by Sophie L

Date: 20260530
Title: Agar Preparation
Purpose: preparation of agar for hydroponics
Procedure :

1. Formula as below:

ddH_2O	10 mL
1/2 MS Medium Powder	0.4 g

2. Add the contents into a beaker.
3. Use hot plate to stir (with a stirring rod) until no sediments are left in the solution.
4. Wrap the neck with aluminium foil and stick a piece of indicator tape on the top.
5. Autoclave for 30 min at 127 °C.
6. Wait for the pressure to reach 0 °C before retrieving the flask.
7. Pipette 800 µL of agar solution into each of the ten tips in the laminar flow.
8. Wait until the agar solution solidifies and place Arabidopsis seeds on all of the tips.
9. Cover them with aluminium foil and place it in the fridge for 2-10 days.

Done by Ryan Wei, Alex

Date: 20260531
Title: Agar Plate Preparation
Purpose: To observe whether the pH of agar has an effect on *Arabidopsis* growth
Procedure:

1. Formula as stated:

ddH_2O	200 mL
1/2 MS Medium Powder	7.9 g

2. Mix the contents in a conical flask.
3. Use a stirring bar to mix evenly until no sediments are left in the flask (on a hot plate).
4. Wrap the neck with aluminum foil & stick a piece of indicator tape on top.
5. Autoclave for 30 min at 127 °C.
6. Wait for the pressure to reach 0 before retrieving the flask.
7. Dispense 30 mL of agar solution into each of the 6 plates in the laminar flow.

Done by Chloe, Professor Kyle, Recorded by Sophie L